

DisasterClaw: A georeferenced imagery benchmark for task-dependent UAV disaster inspection

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Abstract

Aerial disaster inspection poses a task-dependent sensing question: when does a changed view improve the requested answer, and when does it discard necessary context? DisasterClaw turns paired, georeferenced xBD imagery into an interactive inspection benchmark. Each question has a fixed geographic region and annotation-derived answer, while vehicle actions change the rendered observation; the runner records the resulting evidence and answer. The human-reviewed final set contains 160 presence, damage, count, and spatial questions over 44 regions in three events. A development-fitted task-majority baseline reaches 57.50%, while text-only and permitted-runtime-metadata controls reach 28.75% and 36.25%; the image-aware no-dedicated-recheck operating point reaches 72.29%. At 161 executed T1 recheck transitions, a fixed-view matched-call control is correct on 50.31% and the changed-view branch on 59.63% (paired difference +9.32 points; 95% ROI-cluster interval $[-2.00, +21.74]$). The branches contain 36 wrong-to-correct and 21 correct-to-wrong transitions, showing both corrections and harms rather than a uniformly positive effect. Complete-policy results are consistent with this interpretation: T1 answers 360/480 question-repeat instances correctly (75.00%), compared with 347/480 (72.29%) for the search-enabled no-dedicated-recheck arm, whose adjusted interval spans zero. DisasterClaw therefore provides a traceable measurement framework for task-dependent observation value, not a demonstration of a universally superior recheck policy. The benchmark uses resampled orthographic imagery and a fixed perception backbone with prior xBD exposure.

Keywords: UAV disaster inspection, Georeferenced simulation, Task benchmark, Active perception, Closed-loop evaluation, Vision-language agents

1. Introduction

An aerial inspection agent may need to answer whether a building is damaged, how many damaged buildings lie in a region, or where the nearest one is. Descending toward a building can reveal more local detail, but the narrower footprint can hide other buildings needed for a regional answer. The relevant question is therefore whether an additional observation improves the requested assessment, not merely whether an image or a classifier output changes.

Static disaster-image benchmarks hold the image fixed, while aerial navigation benchmarks primarily test movement and target finding. Existing aerial embodied benchmarks already support action-conditioned observations, and remote-sensing agents already support disaster interpretation and structured tool use [1, 2, 3, 4, 5, 6]. The narrower measurement considered here is whether a fixed geographic question and reference answer can be evaluated under a changing observation footprint while retaining a trace from the executed action to geographic evidence and the answer call. Studying the value of another view therefore requires a fixed task region and answer, a controlled change in observation, and a record connecting the action to perception evidence and the final answer. These requirements can be

implemented as image operations, but standardizing them in an episode interface makes the comparison repeatable: a policy request can be distinguished from an executed move, and answer changes can be inspected alongside both view changes and later generation calls.

We present DisasterClaw, an imagery-grounded platform and task benchmark for this question. Paired xBD tiles, building annotations, task regions, and simulated vehicle states are registered to a common map. The benchmark fixes annotation-derived answers on geographic regions while the agent searches or changes its observation footprint. Four task families place different demands on local detail, regional coverage, and geographic reference. A console supports inspection of the scene; the headless runner produces the quantitative episodes.

We use the platform to follow one evidence chain. A development camera ladder first measures how a narrower footprint changes building localization and damage decisions, including both corrections and harms. On the frozen final set of 160 human-reviewed questions from 44 regions in three events, shortcut controls test whether question text or permitted runtime metadata alone can explain task accuracy. A fixed-view matched-call diagnostic then forks

every executed T1 recheck: the fixed branch reruns perception and answering on the old image, whereas the changed branch executes the logged recheck. This design separates a changed observation from merely receiving another answer opportunity at the transition level. Complete-policy comparisons remain operating-point measurements rather than estimates of the isolated effect of a new image.

The contributions are:

1. a georeferenced, ROI-scoped task benchmark in which the reference answer stays fixed as the observation changes, with an explicit boundary between runtime task cues and scoring labels;
2. an inspection interface that records requested and executed actions, rendered views, predicted evidence, answer calls, and terminal outcomes within the same episode; and
3. an empirical characterization of observation value using shortcut controls, fixed-view matched calls, and task-wise help-harm transitions; the accompanying task-conditioned policy is an instantiated operating point rather than the paper’s principal contribution.

The platform renders existing orthographic imagery under simplified motion. Its results concern this controlled observation setting and do not measure physical UAV flight.

2. Related work

2.1. Aerial interaction and embodied evaluation

AerialVLN, CityNav, and OpenFly emphasize language-conditioned aerial navigation, while AeroVerse and BEDI broaden evaluation toward perception, spatial reasoning, planning, motion, and perception–decision–action loops [1, 7, 2, 8, 3]. CityEQA evaluates question answering after active exploration in a realistic urban simulator [4]. These systems establish that actions can change an agent’s observation and that aerial embodied behavior can be evaluated at scale. Several also provide substantially richer 3D simulation, real-world data, or broader embodied tasks than the controlled orthographic setting used here.

2.2. Disaster and remote-sensing agents

xBD supplies paired disaster imagery and building-damage annotations [9]. RescueADI, ThinkGeo, and DisasterM3 study tool use or multimodal disaster interpretation from remote-sensing records [5, 6, 10]. These resources motivate disaster-specific questions, while a static image answer alone cannot reveal whether the agent’s choice to observe again was useful. DisasterClaw binds question scope, vehicle state, rendered observation, and terminal answer in one record. This binding also exposes a failure mode: a view that improves one building prediction can remove evidence required by a count or spatial question.

Table 1 summarizes the evaluation capabilities most relevant to this distinction. The comparison is capability-oriented rather than a ranking of benchmark generality.

Among the compared benchmarks, DisasterClaw explicitly standardizes a fixed annotation-derived geographic task and answer under view change, geographic evidence traces across observations, and a matched fixed-view answer call. Its contribution is this narrower evaluation contract rather than simulator realism or general UAV-agent capability.

2.3. Task value of an observation

Active perception treats sensing as a task-directed action [11]. Entropy and calibrated probabilities can guide such actions, but neither directly measures the correctness of a later task answer [12, 13]. We therefore treat localization, answer correction or harm, executed rechecks, and terminal accuracy as distinct observations. The task-conditioned rule in this paper is a fixed heuristic for descending subject to predicted ROI coverage and motion cost. Its complete-policy evaluation remains an operating-point comparison; we therefore add a fixed-view matched-call diagnostic at executed recheck transitions to control for the additional answer opportunity and isolate the changed-observation/evidence path as far as this transition-level design permits.

3. The DisasterClaw platform

3.1. From geographic records to inspection episodes

DisasterClaw makes a task, an observation, and an answer refer to the same geographic scene (Fig. 1). Paired xBD images are positioned with pixel-to-geographic affine transforms. Building polygons and damage labels define the reference task; a vehicle state and observation window determine what the agent can see. The operator console and headless runner use the same scene and controller modules. All results in this paper come from the headless runner.

The active disaster tile is embedded at its recorded map footprint. Geographic coordinates locate the task ROI, marked building when applicable, vehicle, and rendered window. Disaster imagery supplies the annotated task evidence; a basemap can supply context outside available tiles but contributes no damage ground truth. The task answer remains defined on the ROI when search or recheck changes the observation window. This fixed-answer, variable-view contract is the benchmark capability used in the experiments, rather than the map display itself.

3.2. Controlled observation and action

The nadir renderer uses a 60° horizontal field of view and a 1024-pixel output width. At height h , the ground-footprint width L and nominal ground sampling distance g are

$$L(h) = 2h \tan(\theta/2), \quad g(h) = \frac{L(h)}{W}. \quad (1)$$

The assumed source scale is 0.5 m/pixel, so a 1024-pixel tile defines a nominal 512 m width. Footprints of three, two, and one tile widths give cruise, intermediate, and

Table 1: Capability-oriented comparison of benchmarks most relevant to task-dependent observation value.

Benchmark	Primary task	Fixed geographic				
		Active view change	Disaster-specific	task/answer under view change	Answer after active sensing	Explicit observation-value isolation
AerialVLN	Aerial vision-language navigation	✓	—	—	—	—
OpenFly	Large-scale aerial vision-language navigation	✓	—	—	—	—
BEDI	UAV embodied-agent evaluation	✓	△	—	△	—
CityEQA	Embodied question answering	✓	—	△	✓	—
RescueADI	Disaster remote-sensing agent	—	✓	—	△	—
ThinkGeo	Tool-augmented remote-sensing agent	△	△	—	△	—
DisasterM3	Disaster vision-language benchmark	—	✓	—	—	—
DisasterClaw	Disaster inspection and observation-value measurement	✓	✓	✓	✓	✓

“—” means that the capability is not the primary evaluation contract reported by the work, rather than that it is technically impossible. △ denotes a related but non-equivalent capability. DisasterClaw changes controlled orthographic image footprints rather than physical 3D viewpoints.

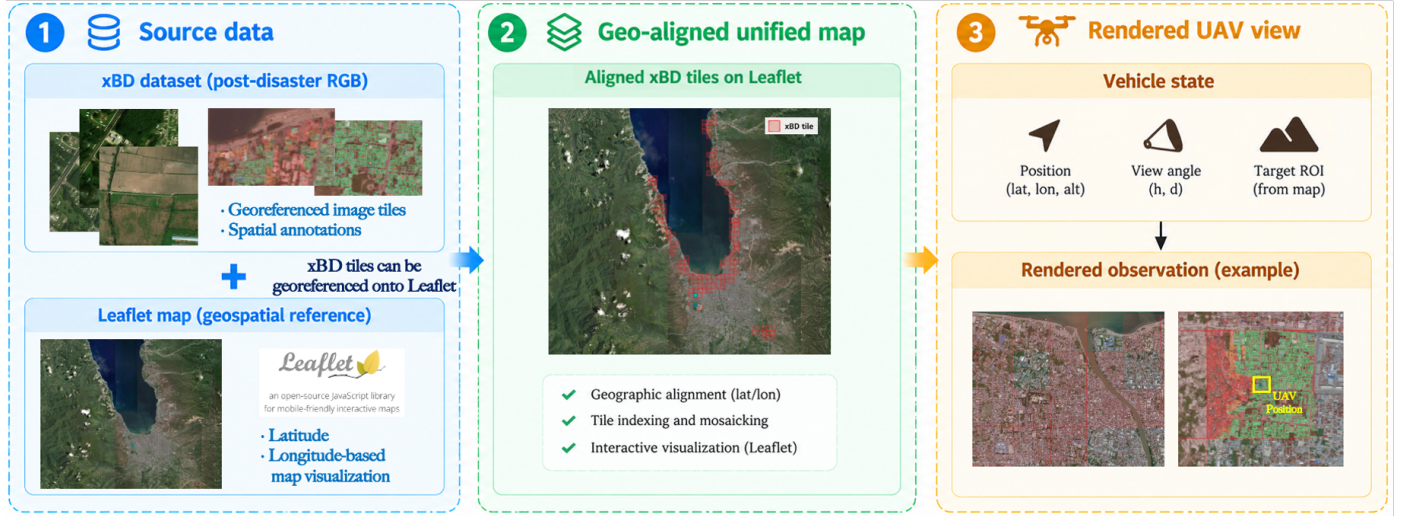


Figure 1: Dataset and scene construction. Georeferenced xBD imagery and spatial annotations are aligned with a Leaflet map, indexed into a unified geographic scene, and rendered according to vehicle state and the target ROI. The basemap supplies geographic context but no damage ground truth.

floor states at approximately 1330.2, 886.8, and 443.4 m, with nominal GSDs of 1.5, 1.0, and 0.5 m/pixel (Fig. 2). Geographic placement comes from the recorded affine transforms; the 0.5 m/pixel value defines this camera ladder and is not a claim that every source tile has an independently verified physical scale.

For paired damage perception, the post-disaster image is sampled at the current footprint and output size; the pre-disaster window is retained at the assumed source scale before both channels enter the detector input geometry. Their preprocessing is asymmetric. Consequently, a camera-ladder response combines changed footprint and context with resampling and detector-scale effects; it cannot be attributed uniquely to additional physical image detail.

The vehicle model stores geographic position, height, heading, and task state. Search can move the vehicle and produce another rendered observation. A dedicated recheck may center on a target and descend, subject to an execution and budget check. The episode log distinguishes requested actions from executed movements, observation identifiers, predicted objects, answer calls, and the final

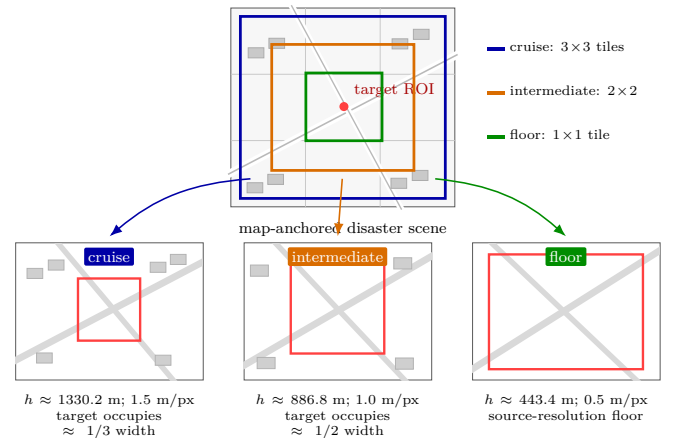


Figure 2: Camera-ladder abstraction. Each state renders 1024×1024 pixels from the same orthographic source; descending narrows the footprint and cannot reveal detail below the assumed source resolution.

normalized answer. An unchanged-view answer mode is implemented for diagnosis, but it is not part of the frozen policy comparison reported here. Motion is simplified position interpolation, and rendering existing orthographic imagery does not create new viewpoints or acquisition times.

4. Task benchmark construction

4.1. Fixed task scope and reference answers

Each question binds a source tile, a geographic ROI, an initial vehicle state, an answer vocabulary, and an annotation-derived reference answer. A building belongs to the reference ROI when its centroid lies inside the ROI bounds. Let B_R be these buildings and $D_R \subseteq B_R$ the buildings labelled damaged. Minor, major, and destroyed xBD labels map to the binary damaged class; unclassified records do not define a binary answer. The reference answer is computed before a policy runs and is unchanged by later crops or movements.

The four task families make the detail-coverage tension testable. Presence asks whether $|D_R| > 0$: one convincing positive may suffice, while a negative requires regional evidence. Damage asks for the binary state of a designated building, whose location is a permitted task cue. Count maps $|D_R|$ to 0, 1, 2, or 3+ and requires aggregation across the ROI. Spatial asks for the eight-sector bearing from the ROI center to the nearest damaged building and requires a nonempty D_R . These are annotation and geometry rules, not language-model-generated reference answers.

Figure 3 illustrates how platform inputs are converted into task outputs. Spatial localization associates predicted damaged buildings with geographic position before computing a relative direction; damage counting aggregates building-level predictions over the visible region; and an active inspection instruction expands into navigation, descent, observation, and reporting actions. These are platform-run examples used to visualize the interface, not samples from the frozen 160-question evaluation set.

4.2. Final-set composition and input boundary

The frozen set has 160 questions over 44 ROIs: 48 in Hurricane Michael, 51 in the Moore tornado, and 61 in Nepal flooding. Table 2 reports the answer counts. The count task is concentrated in 0 and 3+, with only one 1 and no 2; this is a visible class imbalance, so count accuracy alone cannot establish that a policy distinguishes all four buckets. Spatial answers are more dispersed, but that task also requires finding the correct nearest damaged building. Questions from the same ROI share source context and are not independent scenes.

The source task record contains fields needed both at run-time and for scoring. The controller receives the question, answer choices, ROI bounds, initial state, and a marked reference ID and coordinate only for a damage question.

Table 2: Frozen question and answer distribution. Counts refer to distinct questions, before policy runs or generation repeats.

Task	n	Reference answers (count)
Presence	44	yes 30; no 14
Damage	44	damaged 29; no damage 15
Count	43	0: 14; 1: 1; 2: 0; 3+: 28
Spatial	29	N 3; NE 4; E 3; SE 4; S 3; SW 3; W 4; NW 5

It receives detector-predicted boxes, probabilities, and geographic projections after observation. The reference answer, xBD damage labels and polygons, and the true nearest-damaged-building coordinate for a spatial question stay on the construction/scoring side. In particular, the spatial answer cannot be computed from a supplied ground-truth target coordinate in the agent path; the target must be selected from predicted evidence.

Eligible records require usable post-disaster imagery, a geographic footprint, and building labels; pre-disaster imagery supplies the archival damage reference. The generator excludes consumed development tiles, checks answer vocabulary, geometry, duplicate records, and split membership, and retains ambiguity flags for review. The final questions were reviewed by people before freezing. These checks establish the reference construction and input boundary; they do not by themselves show that the questions are visually answerable without shortcuts. Section 7 therefore evaluates text-only and permitted-metadata controls on the same frozen set. The comparison tests whether the complete image-aware task path exceeds controls using these allowed non-image inputs; because the controls also use a modality-specific prompt and omit the hybrid rule path, it is not a pure image-only causal contrast.

5. Task-conditioned observation policy

5.1. Episode state and evidence memory

An episode begins with a question q , geographic ROI, initial state $s_0 = (x_0, y_0, h_0)$, and fixed reference answer y_q . At step t , the renderer produces $I_t = \mathcal{R}(\mathcal{M}, s_t)$; ChangeOS supplies predicted building locations and binary damage probabilities, and the hybrid Qwen2.5-VL-7B-Instruct answer path proposes an answer. The controller can answer, abstain, search, or request a dedicated recheck. The terminal normalized answer is scored against y_q ; abstentions are incorrect for the primary endpoint. Every policy uses the same frozen ChangeOS checkpoint and answer path.

Predicted objects are projected into geographic coordinates. For non-damage tasks, sightings are associated when their predicted centers are within 6 m or their geographic boxes have intersection-over-union at least 0.2; damage sightings use the public marked-building reference ID. Each track retains observation IDs and averages binary probabilities with inverse-GSD weights. Count and spatial answers use tracks detected in the current observation, so a stale historical detection cannot independently supply a current nearest target. This memory can still merge nearby

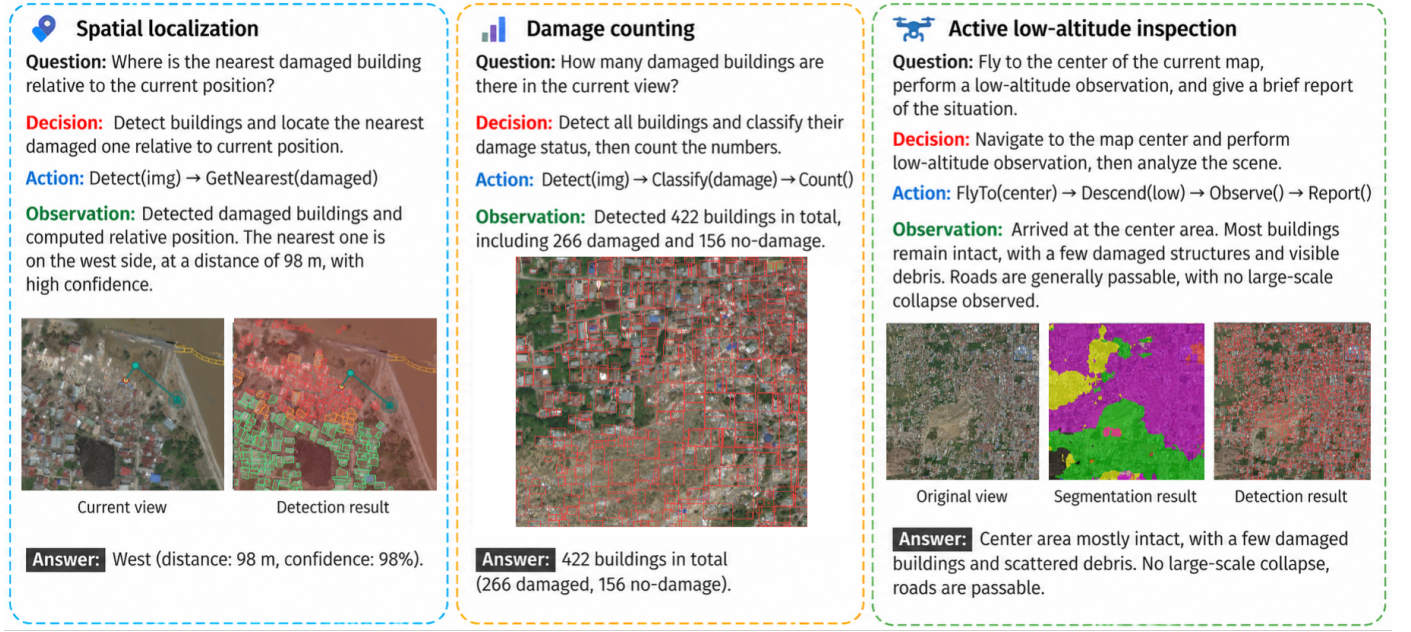


Figure 3: Representative platform-run input-output traces. The examples show spatial localization from mapped detections, aggregation for damage counting, and an active low-altitude inspection sequence. They illustrate the interface and action vocabulary; quantitative results use the separately frozen benchmark and headless runner.

buildings or miss an object, and neither error is removed by high geometric ROI coverage.

5.2. An instantiated recheck gate, not a general view planner

For a candidate recheck, $T1_TASK$ uses

$$U_q = u_q(1 - h_{t+1}/h_t) - 0.05 \hat{T}_{\text{move}}/60 - (1 - \hat{C}_{\text{ROI}}), \quad (2)$$

where $u_q \in [0, 1]$ is a task-specific uncertainty score, $\hat{T}_{\text{move}}$ is configured-speed motion time in seconds, and $\hat{C}_{\text{ROI}}$ is predicted coverage after the move. A candidate executes only when $u_q \geq 0.5$, $U_q \geq 0.05$, and, for count or spatial questions, predicted ROI coverage is at least 0.98. The positive utility term comes only from descent; this rule is a task- and coverage-constrained descent trigger, not a general value function for same-height viewpoint changes or a calibrated expected accuracy gain.

The uncertainty score follows the requested answer. For damage, it is normalized binary entropy for the marked predicted building; a visible marked location with no matched detection receives $u_q = 1$. For count, a capped Poisson-binomial distribution from detected buildings' damage probabilities gives entropy over 0, 1, 2, 3+. For a matched spatial candidate, the score is the maximum of its binary entropy and a GSD-dependent estimate of proximity to an eight-sector bearing boundary. Presence is answered from the wide view when possible; missing evidence enters search rather than a local recheck. Count uncertainty conditions on detected objects and does not model missed buildings. The spatial score uses a selected predicted target and does

not fully represent uncertainty about which damaged building is nearest.

Damage rechecks may move toward the marked building by at most 40 m horizontally and descend. Count and spatial rechecks keep the geographic center and descend only while the task ROI remains covered. Each active policy has a cap of two dedicated rechecks. The stored policy identifier $A0_HOLD$ is referred to here as $A0_NO_RECHECK$: it has a zero dedicated-recheck cap but, like the other policies, may use up to six search steps. Search may move and render a new image. Thus $A0_NO_RECHECK$ is not a fixed-view control, and the reported recheck count is not total movement, rendering, or model-call cost.

The primary comparators are $A1_RANDOM$ (a seeded 0.5 recheck trigger), $A2_ALWAYS$ (request a feasible recheck), and $A3U_RAW_ENTROPY$ (uncalibrated binary-entropy trigger). $A3_ENTROPY$ and $A4_CONFORMAL$ are supplemental policies. All seven configurations have the same question set and search cap, while their realized actions and later answer-call counts may differ. This comparison therefore measures whole-policy outcomes at their realized operating points.

6. Evaluation protocol

The evaluation follows the evidence chain from changed view to building prediction to task answer. The development camera-ladder diagnostic compares cruise, intermediate, and floor observations for the same 7,238 annotated buildings in 43 ROIs. Its fixed-denominator measure treats an unmatched ground-truth building as incorrect; matched-

only classification describes a selected subset and is not used as the task endpoint. This diagnostic helps interpret whether the rendered view changes available predicted evidence.

The final online set is separate from the 160-question development set used to select policy rules. Seven frozen policies each answer the same 160 final questions under three generation repeats, yielding $7 \times 160 \times 3 = 3,360$ episodes. The repeats change seeded generation conditions for the same questions, not the number of independent questions, ROIs, or events. The primary endpoint is terminal question accuracy, with abstentions counted incorrect. Executed dedicated rechecks, within-policy answer transitions, task type, and event are descriptive companion measures.

We audited spatial separation beyond distinct tile IDs. The 43 development ROIs and 44 final ROIs share no tile ID or xBD building UID (0 shared among 7,238 and 4,963 eligible ROI buildings), but 9/1,892 cross-set ROI pairs intersect; the maximum ROI IoU is 0.461 and the maximum fraction of the smaller ROI covered by the other is 0.631. Comparing 129 centered development camera-ladder footprints with all 802 recorded T1 final observation footprints across three repeats, 1,871/103,458 cross-set pairs intersect. The maximum footprint IoU is 0.882, and one smaller footprint lies entirely inside a larger one (overlap fraction 1.000). These are geographic-window overlaps, not evidence that the same xBD pixels or physical building instances were used in both sets; a different tile can carry a different UID for the same physical structure. Thus question and tile disjointness does not establish spatial independence.

We add two controls on the frozen final set. First, the task-majority baseline predicts one answer per task type using a question/ROI-disjoint development set. S1_LANGUAGE receives question text, choices, and task type; S2_METADATA additionally receives runtime-visible ROI bounds/tile, initial pose, and, for damage questions, the permitted marked-building reference and coordinate. Both exclude images, detector outputs, annotation labels and polygons, the reference answer, and the spatial true-nearest target. S1 and S2 use the same Qwen checkpoint, answer vocabulary, generation settings, seed schedule, and exact-match scoring as the main answer model, but a modality-appropriate text-only JSON prompt and no hybrid rule path.

Second, a fixed-view matched-call diagnostic is run at every one of the 161 executed T1 dedicated rechecks. Immediately before the live recheck, the fixed branch clones the semantic map and evidence memory, remains at the old pose, reruns perception on the old view, and calls the ordinary hybrid answer path. The changed branch executes the logged action, perceives the new view, and calls that same answer path. The two branches share question, repeat, next-step seed, model, and generation settings; their prompts differ only through the resulting predicted evidence. A strict gate requires a byte-identical fixed-branch

marked image and exactly one fixed and changed model call with the shared seed. This is a paired candidate-answer transition diagnostic, not a forked terminal-episode comparison, and the fixed answer is never fed into T1 control.

Four T1_TASK contrasts were specified before the final run: against A0_NO_RECHECK (internal identifier A0_HOLD), A1_RANDOM, A2_ALWAYS, and A3U_RAW_ENTROPY. Comparisons pair the same question and repeat. The 5,000-draw bootstrap resamples ROI clusters, retaining their questions and repeats, and reports Bonferroni-adjusted 98.75% intervals for the four contrasts. The intervals characterize uncertainty among the observed 44 ROI clusters; three events are insufficient to infer transfer to unseen disasters. Task and event breakdowns are descriptive, not additional adjusted tests.

For the matched diagnostic, correctness pairs and answer disagreement are summarized overall and by task, with a 5,000-draw ROI-cluster bootstrap 95% interval for the changed-minus-fixed difference. Shortcut results report overall and task-wise accuracy; image-aware A0_NO_RECHECK and T1 gaps to S1/S2 are paired by question and repeat with the same ROI-cluster bootstrap. These controls are diagnostic checks of visual dependence and observation value; they are not additional policy-selection tests.

The primary cost count records executed dedicated rechecks. Search can also move and render, and policies can make different numbers of later Qwen calls; total movement, rendering, perception, and generation costs are therefore not summarized by the recheck column. Episode wall time is measured software runtime, not flight duration. The final protocol identifies the frozen task, model, prompt, and run configurations; its audit and release boundary are summarized in Appendix A. A runnable single-episode scene and scoring replay is provided in Appendix B.

7. Results and discussion

7.1. Benchmark validity: image-aware paths exceed shortcut controls

The final answer distribution has a substantial task-conditioned majority baseline: the development-fitted task-majority rule attains 57.50% (92/160). This number is driven by the frozen answer distribution, especially the count task, rather than by visual interpretation. The text-only S1_LANGUAGE control reaches 28.75% over 480 question-repeat instances, and the permitted-metadata S2_METADATA control reaches 36.25% (Table 3). Neither control receives imagery or detector evidence. In contrast, the image-aware A0_NO_RECHECK and T1_TASK operating points reach 72.29% and 75.00%, respectively. Their paired gaps to S1 are +43.54 and +46.25 points, and to S2 are +36.04 and +38.75 points; all four 95% ROI-cluster intervals exclude zero. The complete image-aware task paths therefore exceed controls based on question wording and permitted non-image metadata. Because S1/S2 also

use a modality-specific prompt and omit the hybrid rule path, this is a pipeline-level benchmark control rather than a pure image-only causal estimate.

7.2. Changed observations beyond another answer call

The policy comparison alone cannot distinguish a new image from another answer opportunity. We therefore forked all 161 executed T1 rechecks from the same pre-recheck state. In the fixed branch, the old view was perceived and answered again; in the changed branch, the original recheck was executed before perception and answering. The strict validity gate accepted all 161 pairs, with no missing pair or execution error. The branch schedule reproduced the original recheck-step locations in 118 of 119 episodes; the one mismatch retained the same two rechecks but placed the second at a different step, reflecting re-runnable perception-path variation.

Changed-view candidate answers are correct in 96/161 transitions (59.63%), versus 81/161 (50.31%) for the fixed-view matched call, a paired difference of +9.32 points (95% ROI-cluster interval $[-2.00, +21.74]$). The point estimate is accompanied by 36 fixed-wrong/changed-correct transitions and 21 fixed-correct/changed-wrong transitions; 44 pairs are wrong in both branches and 60 correct in both. Answers disagree in 42.24% of pairs. Thus the changed observation can alter the answer path beyond a repeated call on the old image, but the clustered interval does not establish a stable positive average effect in the sampled ROIs.

Excluding the one episode whose rerun placed its second recheck at step 2 rather than reference step 3 removes two wrong-wrong pairs. The remaining 159 pairs from 118 episodes have 81/159 fixed-view and 96/159 changed-view correct candidates, a +9.43-point paired difference (95% ROI-cluster interval $[-2.03, +22.00]$), with the same 36 help and 21 harm transitions. As a separate old-view repeat check, the pre-recheck and fixed-branch image bytes and normalized candidate answers agree in all 161/161 original forks (159/159 after exclusion). The two old-view calls use different step seeds and evidence prompts, however, so this is observed answer stability under identical image bytes rather than a controlled same-prompt, same-seed repeatability estimate.

Figure 4 grounds the aggregate transitions in two frozen T1 episodes. In the damage episode `hurricane-michael_00000435_post_disaster__damage_0001_9430`, successive observations at approximately 1330, 887, and 443 m increase the marked building’s predicted damage probability from 0.158 to 0.369 and 0.588. The answer consequently changes from the incorrect “no damage” to the correct “damaged.” This is the intended local-detail benefit of a recheck.

The spatial episode `hurricane-michael_00000306_post_disaster__spatial_0092_5452` shows the complementary risk. The wide observation selects a damaged building north of the ROI center and answers correctly. After descent, the evidence ranking selects a different building southwest of the center, changing a correct answer to

an incorrect one. A further descent is rejected because predicted ROI coverage is 0.340, below the 0.98 gate. The failure is therefore associated with a target-identity switch under reduced regional context, not merely with noise in the bearing of a fixed target.

7.3. Changed observations create opportunity and risk

The camera-ladder diagnostic holds 7,238 annotated buildings in 43 development ROIs fixed while changing the rendered footprint. ChangeOS localization recall rises from 22.09% at cruise to 34.06% at intermediate and 56.52% at floor. With every unmatched ground-truth building counted as incorrect, binary building accuracy rises from 20.01% to 31.06% and 53.65%. Among matched buildings only, binary damage accuracy is 90.56%, 91.20%, and 94.92%, while matched-only macro-F1 is 0.8714, 0.8743, and 0.9232. The paired cruise-to-floor outcomes include 2,696 incorrect-to-correct changes and 261 correct-to-incorrect changes. This building-level diagnostic establishes that changed rendering changes available predicted evidence, but asymmetric pre-/post-image processing and changed context prevent attributing the response uniquely to GSD or physical descent.

Table 5 separates localization from conditional damage classification. Localization precision falls to 68.65% at floor even as recall and matched-building damage accuracy rise; unmatched predictions increase to 43.44 per ROI. TP/FP/FN use a one-to-one nearest-centroid match within 12 m, so an unmatched prediction can reflect a duplicate, localization error, or missing annotation as well as a spurious building. The all-GT accuracy denominator includes missed buildings but does not penalize extra predictions; matched-only accuracy and macro-F1 exclude misses and unmatched predictions. Thus the improving damage accuracies and increasing localization FP count describe different parts of the same detection path, without implying that either metric is incorrect.

The matched-call result carries the same non-monotonic pattern to task answers. Damage and count transitions have positive point differences, whereas spatial transitions have a negative point difference (Table 4). A more local view can improve the evidence for a marked building or a damaged-count decision, yet it can change the selected target for a nearest-building direction judgment. These task-wise estimates are descriptive: the recheck subsets are selected by T1 and have only 18 count and 20 spatial transitions.

7.4. Policy operating points remain secondary evidence

Table 6 reports the prespecified complete-policy runs. T1_TASK has the highest observed terminal accuracy, 360/480 question-repeat instances (75.00%), with 161 executed dedicated rechecks. A0_NO_RECHECK (the internal configuration identifier is A0_HOLD) forbids dedicated rechecks but permits search and answers 347/480 (72.29%). A2_ALWAYS makes 923 rechecks and answers 307/480

Table 3: Shortcut controls and image-aware operating points on the frozen final set. S0 uses a development-fitted task majority and is deterministic ($n = 160$); all other rows use three aligned repeats ($n = 480$). S1 receives question text, choices, and task type; S2 additionally receives permitted runtime metadata.

Method	Image	Detector evidence	Runtime metadata	Presence	Damage	Count	Spatial	Overall
S0 task majority	no	no	task only	68.18	65.91	65.12	17.24	57.50
S1 language only	no	no	question only	31.82	34.09	32.56	10.34	28.75
S2 metadata only	no	no	permitted	31.82	58.33	32.56	14.94	36.25
A0_NO_RECHECK	yes	yes	permitted	93.18	66.67	75.97	43.68	72.29
T1_TASK	yes	yes	permitted	93.18	75.76	78.29	41.38	75.00

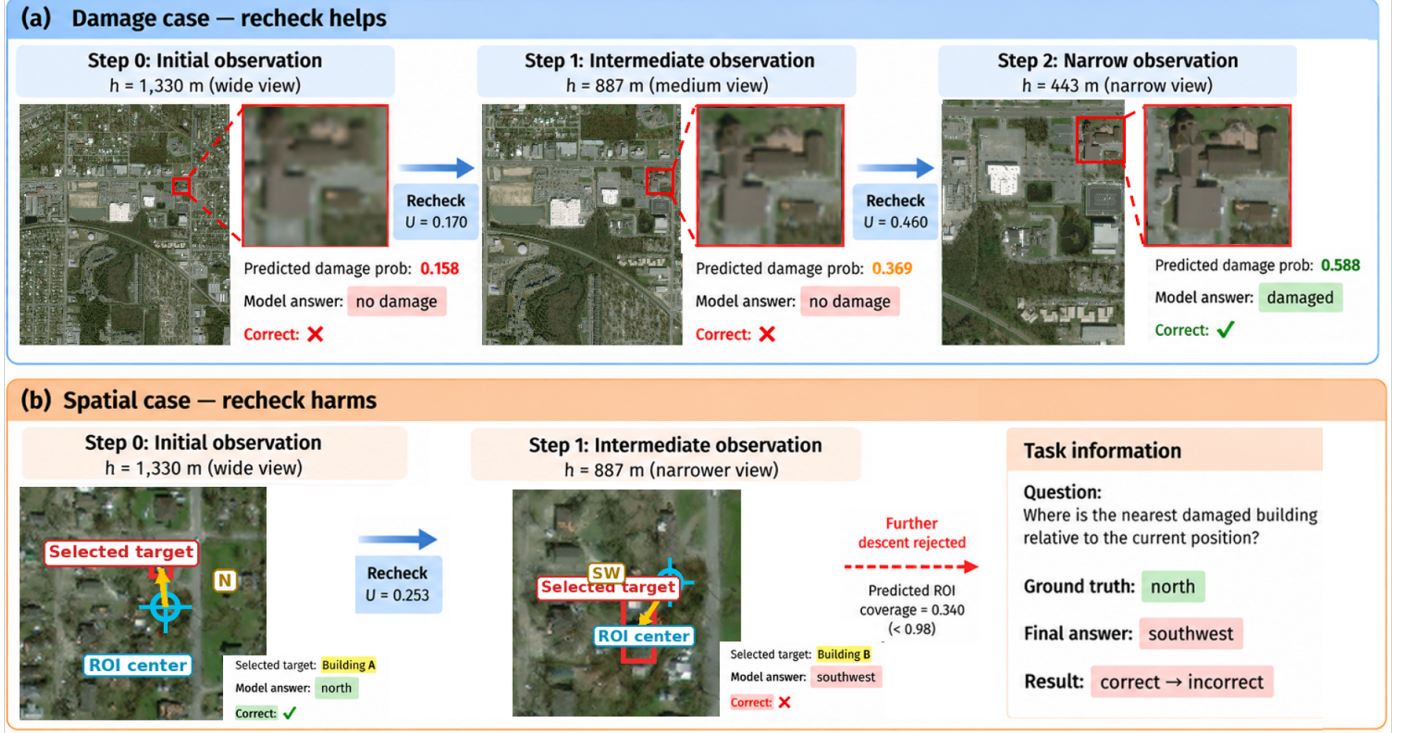


Figure 4: Frozen examples of recheck help and harm. (a) Narrower observations strengthen evidence for a marked building and correct a damage answer. (b) A spatial recheck changes the selected nearest damaged building from the correct northern target to a southwestern target; the coverage gate then rejects further descent. The examples expose the action–observation–evidence–answer chain behind the aggregate help and harm counts.

Table 4: Fixed-view matched-call diagnostic at executed T1 recheck transitions. Candidate-answer correctness is measured immediately after the branch, not at episode termination. Presence has no T1 dedicated recheck.

Task	n	Fixed	Changed	Help	Harm	Δ pp
Damage	123	46.34	58.54	27	12	+12.20
Count	18	83.33	100.00	3	0	+16.67
Spatial	20	45.00	30.00	6	9	-15.00
Overall	161	50.31	59.63	36	21	+9.32

(63.96%). These are realized operating points under a common maximum of two dedicated rechecks for active policies, not matched action, model-call, or total-cost comparisons.

The 2.71-point T1_TASK advantage over A0_NO_RECHECK is an observed difference of 13 correct question-repeat instances, not 13 distinct questions. Its 98.75% multiplicity-adjusted interval includes zero (Table 7), as does the comparison with raw entropy. The positive interval against A2_ALWAYS supports a difference between these complete policies on the sampled ROIs; it does not show that T1’s individual changed observations caused that difference.

At the terminal-episode level, presence is unchanged between T1 and A0_NO_RECHECK, damage and count are numerically higher for T1, and spatial accuracy is

Table 5: ChangeOS localization and binary-damage audit on the same 43 development ROIs and 7,238 annotated buildings at each height. Loc. P/R are localization precision/recall; FP is an unmatched prediction, FN an unmatched GT building, and FP/ROI is FP/43. All-GT accuracy counts unmatched GT as wrong but does not count unmatched predictions; matched accuracy and macro-F1 condition on matched GT only.

View (height)	TP	FP	FN	Loc. P (%)	Loc. R (%)	FP/ROI	All-GT acc. (%)	Matched acc. (%)	Matched macro-F1
Cruise (1330 m)	1,599	502	5,639	76.11	22.09	11.67	20.01	90.56	0.8714
Intermediate (887 m)	2,465	677	4,773	78.45	34.06	15.74	31.06	91.20	0.8743
Floor (443 m)	4,091	1,868	3,147	68.65	56.52	43.44	53.65	94.92	0.9232

Table 6: Primary frozen policy operating points on the same 160 questions under three generation repeats. Rechecks are executed dedicated actions and exclude search.

Policy	Correct / 480	Accuracy (%)	Rechecks
A0_NO_RECHECK	347	72.29	0
A1_RANDOM	326	67.92	419
A2_ALWAYS	307	63.96	923
A3U_RAW_ENTROPY	339	70.63	626
T1_TASK	360	75.00	161

Table 7: Prespecified paired T1_TASK terminal-accuracy contrasts in percentage points. The 98.75% ROI-cluster intervals adjust for four comparisons.

Contrast	Difference	98.75% interval
T1 minus A0_NO_RECHECK	+2.71	[−2.04, +7.50]
T1 minus A1_RANDOM	+7.08	[+0.01, +14.69]
T1 minus A2_ALWAYS	+11.04	[+0.86, +21.38]
T1 minus A3U_RAW_ENTROPY	+4.38	[−4.33, +12.96]

lower (Table 3). Event differences also prevent a general policy claim: T1 versus A0 is 54.86% versus 58.33% in Hurricane Michael, 84.31% versus 77.12% in Moore, and 83.06% versus 79.23% in Nepal. Together with the matched-call control, the interval for the A0 contrast, and incomplete cost observability, these results support the benchmark’s ability to expose task-dependent corrections and harms rather than a generally superior recheck algorithm.

8. Limitations

The benchmark has only 44 ROIs in three events. Questions from nearby regions may share image context, and the fixed ChangeOS checkpoint has prior xBD exposure; the present results cannot establish unseen-disaster transfer. The final answer distribution is uneven, especially for count. The shortcut controls show a substantial image-aware gap on this set, but they do not prove that every question requires visual evidence or rule out all unmeasured shortcuts. Human review and deterministic annotation rules support reference construction rather than a claim of universal visual answerability.

The camera ladder resamples orthographic imagery and uses asymmetric pre-/post-disaster processing. It cannot represent a physical descent’s new angle, occlusion, acquisition conditions, or motion dynamics. The spatial uncertainty score also conditions on one selected predicted target and does not fully represent uncertainty over which

damaged building is nearest; the target switch in Fig. 4 illustrates this unresolved source of harm. A0_NO_RECHECK still permits search, and the policies can differ in subsequent answer calls. The matched-call diagnostic addresses repeated answering at recheck transitions, but it is not a terminal-episode counterfactual; perception may remain nondeterministic even when the fixed pixels are identical, and its 95% ROI-cluster interval crosses zero. The old-view agreement check changes prompts and seeds, so it does not rule out model variance under a strictly identical call. Development and final questions use distinct tiles and UUIDs, but their ROIs and rendered footprints can overlap geographically; this limits claims of spatial independence. Recheck counts exclude search and do not measure total render, perception, generation, or flight cost. Of 3,360 episodes, 42 lack a recorded state after the terminal fly action; this affects complete-motion summaries, not the terminal-answer denominator. The qualitative examples explain observed mechanisms but do not establish that rechecking has a generally positive effect.

9. Conclusion

DisasterClaw fixes the geographic scope and answer of a disaster-inspection question while allowing the observation to change. Its episode record connects actions, predicted building evidence, answer calls, and terminal task outcomes. The four task families expose a tension between local detail and regional coverage that a building classifier score alone cannot show.

On the frozen final set, text-only and permitted-metadata shortcuts remain well below image-aware operating points. At matched recheck transitions, changed observations produce both more wrong-to-correct than correct-to-wrong answers and a positive point estimate over a fixed-view matched call, while the ROI-cluster interval still includes zero. Task-wise patterns are consistent with a local-detail versus regional-coverage trade-off: damage and count transitions improve numerically, whereas spatial transitions worsen. T1_TASK has the highest observed complete-policy accuracy, but its adjusted contrast with the search-enabled A0_NO_RECHECK arm spans zero. The contribution is therefore a benchmark interface and a bounded, traceable measurement of observation value, rather than a demonstrated universal observation policy.

Declarations for author completion

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Data and code availability. We will release the DisasterClaw code, benchmark task definitions, frozen protocols, audited episode traces, analysis scripts, and a runnable minimal reproduction package with its example xBD tile and labels on GitHub. The repository will provide environment instructions, file checksums, and instructions for obtaining the full xBD imagery and model checkpoints from their respective sources. The GitHub URL will be added to the final published version.

Generative AI assistance. Generative AI tools were used to assist manuscript restructuring, English-language editing, reference checking, and preparation of analysis and typesetting scripts. All scientific claims, experimental design, reported results, and final manuscript content were reviewed and approved by the authors, who take full responsibility for the work.

Appendix A. Protocol and provenance

The frozen execution protocol identifies the final task set and consolidated human-review report, three-event manifest, ChangeOS checkpoint, source fingerprint, prompt, policy configuration, and analysis scripts by SHA-256. The planned GitHub release will include the protocol, task and scene manifests, dependency specifications, replay instructions, and verified links to the original imagery and checkpoint sources.

The final run consists of 12 shards with 3,360 question-policy-repeat rows. The shard audit reports complete question and configuration coverage and zero execution errors. This checks execution consistency, not independent reconstruction of the full environment. A separate cost audit finds that 42 episodes end with a fly action and no subsequent recorded state; complete-motion summaries therefore use 3,318 episodes. The terminal-answer analysis retains all 3,360. Search-action detail and component-level render, perception, controller, and Qwen costs are not available, so the action-cost comparison is restricted to executed dedicated rechecks and observed episode wall time.

The supplementary-control report aggregates 12 matched-view shards (161 valid forks, no missing pairs or execution errors) and four shortcut shards (960 valid text-only or metadata-only predictions, no call errors or invalid outputs). The planned GitHub release will include the machine-readable report, task-wise counts, ROI-cluster intervals, and protocol notes describing the fixed-view branch,

allowed shortcut fields, and diagnostic boundaries. It will also include the matched-view and shortcut-control runners and the report-generation script so these results can be regenerated from the released inputs.

For the camera-ladder diagnostic, the construction boundary is summarized below. The table is a reproducibility description, not an ablation of preprocessing choices.

Table A.1: Camera-ladder preprocessing construction used by the diagnostic.

Stage	Pre-disaster	Post-disaster
Geographic crop	matched window	current footprint
Scale handling	assumed source scale	footprint-dependent sampling
Alignment	paired affine placement	paired affine placement
Detector input	retained source-scale channel	resized current view

Appendix B. Minimal frozen episode reproduction

The planned GitHub release will include a minimal example that replays one frozen T1_TASK damage episode, `hurricane-michael_00000435_post_disaster__damage_0001_9430`, generation repeat 0. The example contains the paired 1024×1024 xBD tile images, the post-disaster polygon/UID annotations, a scene manifest with geographic ROI and pixel-to-geographic transform, a compact record of the episode’s observations and actions, a standard-library Python replay script, and exact expected output. SHA-256 digests in the manifest detect asset substitution. This example can be run without detector or language-model weights; it reproduces scene linkage, observation-window geometry, the archived action trace, and reference-answer scoring, but does not rerun perception or generation.

The environment for this replay is Python 3.11 with no third-party packages. The GitHub README will specify how to run the replay script and compare its output with the expected result. For fresh model inference, the release will provide full environment specifications, the frozen task and review manifests, and instructions for obtaining the ChangeOS and Qwen2.5-VL-7B-Instruct checkpoints. A fresh inference run need not reproduce the archived candidate sequence exactly.

The replay first verifies the two image hashes and their dimensions, locates the marked building by xBD UID in the post-disaster label, and computes its binary damage reference from the original `major-damage` subtype. It then constructs each observation footprint from the recorded pose using the 60° field of view and the same geographic conversion used by the renderer. Table B.1 gives the compact recorded action sequence. The final **Report** answer is compared with the annotation-derived reference, yielding the expected output `reference=damaged`, `answer=damaged`, and `correct=true` (the underlying task vocabulary is Chinese).

Table B.1: Expected trace of the single frozen damage episode. Candidate answers are shown in English for readability; the replay emits the original Chinese labels.

Step	Height (m)	Candidate	Action
0	1330.22	no damage	Fly/reobserve
1	886.82	no damage	Fly/reobserve
2	443.42	damaged	Report

The example identifies its source frozen record by question ID, generation repeat, and shard.

The GitHub release will also include a supplementary audit script and its machine-readable output. The script reads the matched-call shards, development camera-ladder report, both frozen question manifests, xBD labels, and final T1 traces. Its output contains the schedule-exclusion calculation, old-view answer agreement, building-UID and geographic overlap audit, and the TP/FP/FN counts in Table 5. No new model inference is needed for these audit calculations.

References

- [1] S. Liu, H. Zhang, Y. Qi, P. Wang, Y. Zhang, Q. Wu, AerialVLN: Vision-and-language navigation for UAVs, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 15384–15394.
- [2] Y. Gao, C. Li, Z. You, J. Liu, Z. Li, P. Chen, Q. Chen, Z. Tang, L. Wang, P. Yang, Y. Tang, Y. Tang, S. Liang, S. Zhu, Z. Xiong, Y. Su, X. Ye, J. Li, Y. Ding, D. Wang, X. Li, Z. Wang, B. Zhao, OpenFly: A comprehensive platform for aerial vision-language navigation, arXiv preprint arXiv:2502.18041v7 (2026). doi:10.48550/arXiv.2502.18041.
- [3] M. Guo, et al., Bedi: A comprehensive benchmark for evaluating embodied agents on uavs, ISPRS Journal of Photogrammetry and Remote Sensing 232 (2026) 910–936. doi:10.1016/j.isprsjprs.2026.01.013.
- [4] Y. Zhao, K. Xu, Z. Zhu, Y. Hu, Z. Zheng, Y. Chen, Y. Ji, C. Gao, Y. Li, J. Huang, CityEQA: A hierarchical LLM agent on embodied question answering benchmark in city space, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, 2025, pp. 12465–12480. doi:10.18653/v1/2025.emnlp-main.630.
- [5] Z. Liu, D. Zhao, B. Yuan, RescueADI: Adaptive disaster interpretation in remote sensing images with autonomous agents, arXiv preprint arXiv:2410.13384 (2024). doi:10.48550/arXiv.2410.13384.
- [6] A. Shabbir, M. A. Munir, A. Dudhane, M. U. Sheikh, M. H. Khan, P. Fraccaro, J. B. Moreno, F. S. Khan, S. Khan, Thinkgeo: Evaluating tool-augmented agents for remote sensing tasks, arXiv preprint arXiv:2505.23752 (2025). arXiv:2505.23752.
- [7] J. Lee, T. Miyanishi, S. Kurita, K. Sakamoto, D. Azuma, Y. Matsuo, N. Inoue, CityNav: A large-scale dataset for real-world aerial navigation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 5912–5922.
- [8] F. Yao, et al., AeroVerse: Uav-agent benchmark suite for simulating, pre-training, finetuning, and evaluating aerospace embodied world models, arXiv preprint arXiv:2408.15511 (2024). arXiv:2408.15511.
- [9] R. Gupta, R. Hosfelt, S. Sajeed, N. Patel, B. Goodman, J. Doshi, E. Heim, H. Choset, M. Gaston, xBD: A dataset for assessing building damage from satellite imagery, arXiv preprint arXiv:1911.09296 (2019). doi:10.48550/arXiv.1911.09296.
- [10] J. Wang, et al., Disaster3: A remote sensing vision-language dataset for disaster damage assessment and response, Advances in Neural Information Processing Systems (2025). arXiv:2505.21089.
- [11] R. Bajcsy, Active perception, Proceedings of the IEEE 76 (8) (1988) 996–1005.
- [12] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning, Vol. 70 of Proceedings of Machine Learning Research, 2017, pp. 1321–1330.
- [13] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon, B. Lakshminarayanan, J. Snoek, Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift, in: Advances in Neural Information Processing Systems, Vol. 32, 2019.